$$\bar{d} = \frac{1}{N}\sum_{i=1}^{N} d_i$$
$$\bar{d} = \frac{1}{12}\sum_{i=1}^{12} d_i$$
$$\bar{d} = \frac{1}{12} * 23.94mm$$
$$\bar{d} = 1.995mm$$
$$----$$
$$u_A(d) = \sqrt{\frac{1}{N(N-1)}\sum_{i=1}^{N}(d_i - \bar{d})^2}$$
$$u_A(d) = \sqrt{\frac{1}{12(12-1)}\sum_{i=1}^{12}(d_i - 1.995)^2}$$
$$u_A(d) = \sqrt{\frac{1}{132} * 0.0003}$$
$$u_A(d) = 0.001414mm$$
$$----$$
$$u_A(d) = \frac{0.01}{\sqrt{12}}$$
$$u_B(d) = 0.002887mm$$
$$----$$
$$u_C(d) = \sqrt{u_A^2(d) + u_B^2(d)}$$
$$u_C(d) = \sqrt{0.001414^2 + 0.002887^2}$$
$$u_C(d) = 0.00321468mm$$
$$----$$
$$\bar{r} = \frac{\bar{d}}{2}$$
$$\bar{r} = \frac{1.995mm}{2}$$
$$\bar{r} = 0.9975mm$$
$$----$$
$$u(r) = \sqrt{\frac{\partial r}{\partial d}^2 u_C^2(d)}$$
$$u(r) = \sqrt{\frac{1}{2}^2 * 0.00321468^2}$$
$$u(r) = 0.00160734mm$$
$$----$$
$$\bar{m} = \frac{1}{12}(m_{celkem} - m_{miska})$$
$$\bar{m} = \frac{1}{12}(22.4593g - 22.0744g)$$
$$\bar{m} = 0.032075g$$
$$----$$
$$u(m) = \frac{0.0001}{\sqrt{12}}$$
$$u(m) = 0.0000288675g$$
$$----$$
$$\bar{\rho_k} = \frac{\bar{m}}{\frac{4}{3}\pi\bar{r}^3}$$
$$\bar{\rho_k} = \frac{0.032075g}{\frac{4}{3}\pi 0.9975mm^3}$$
$$\bar{\rho_k} = 0.007715g * mm^{-3} = 7715kg * m^{-3}$$
$$u(\rho_k) = \sqrt{(\frac{\partial \rho_k}{\partial r})^2 u^2(r) + (\frac{\partial \rho_k}{\partial m})^2 u^2(m)}$$
$$u(\rho_k) = \sqrt{(\frac{-9\bar{m}}{4\pi\bar{r}^4})^2 u^2(r) + (\frac{3}{4\pi\bar{r}^3})^2 u^2(m)}$$
$$u(\rho_k) = \sqrt{(\frac{-9 * 0.032075g}{4\pi * 0.9975mm^4})^2 * 0.00160734mm^2 + (\frac{3}{4\pi * 0.9975mm^3})^2 * 0.0000288675g^2}$$
$$u(\rho_k) = 0.0000379363g * mm^{-3} = 37.9363kg * m^{-3}$$
$$----$$

$$u_B(\Delta L) = \frac{0.001}{\sqrt{12}}$$
$$u_B(\Delta L) = 0.000289m$$
$$----$$
$$u(\Delta L) = \sqrt{0.002^2 m + 0.000289^2 m}$$
$$u(\Delta L) = 0.00202077m$$
$$----$$
$$u(\rho) = \frac{5}{\sqrt{12}}$$
$$u(\rho) = 1.443376kg * m^{-3}$$
$$----$$
$$\bar{T} = \frac{1}{N}\sum_{i=1}^{N} T_i$$
$$\bar{T} = \frac{1}{8}\sum_{i=1}^{8} T_i$$
$$\bar{T} = \frac{1}{8} * 287.9$$
$$\bar{T} = 35.9875s$$
$$----$$
$$u_A T = \sqrt{\frac{1}{N(N-1)}\sum_{i=1}^{N}(T_i - \bar{T})^2}$$
$$u_A T = \sqrt{\frac{1}{8(8-1)}\sum_{i=1}^{8}(T_i - 35.9875)^2}$$
$$u_A T = \sqrt{\frac{1}{8(8-1)}0.016950}$$
$$u_A T = 0.0173977s$$
$$----$$
$$u_B T = \frac{0.01}{\sqrt{12}}$$
$$u_B T = 0.002887s$$
$$----$$
$$uT = \sqrt{u_A^2 T + u_B^2 T + T_{reakce}^2}$$
$$uT = \sqrt{0.0173977^2 + 0.002887^2 + 0.169^2}$$
$$uT = 0.169918s$$

$$\bar{\eta} = \frac{2}{9} * g * r^2 * (\rho_k - \rho) * \frac{\Delta T}{\Delta L}$$
$$\bar{\eta} = \frac{2}{9} * 9.81 * 0.0009975^2 * (7715 - 965) * \frac{35.9875}{0.55}$$
$$\bar{\eta} = 0.958021 N * s * m^{-2}$$
$$----$$
$$u(\eta) = \sqrt{(\frac{\partial \eta}{\partial r})^2 u^2(r) + (\frac{\partial \eta}{\partial \rho_k})^2 u^2(\rho_k) + (\frac{\partial \eta}{\partial \rho})^2 u^2(\rho) + (\frac{\partial \eta}{\partial \Delta T})^2 u^2(\Delta T) + (\frac{\partial \eta}{\partial \Delta L})^2 u^2(\Delta L)}$$
$$u(\eta) = \sqrt{\begin{array}{l}(\frac{4gr\Delta T(\rho_k - \rho)}{9\Delta L})^2 u^2(r) + (\frac{2gr^2\Delta T}{9\Delta L})^2 u^2(\rho_k) + (\frac{2gr^2\Delta T}{9\Delta L})^2 u^2(\rho) + \\ (\frac{2gr^2(\rho_k - \rho)}{9\Delta L})^2 u^2(\Delta T) + (\frac{-2gr^2\Delta T(\rho_k - \rho)}{9\Delta L^2})^2 u^2(\Delta L)\end{array}}$$
$$u(\eta) = \sqrt{\begin{array}{l}(\frac{4 * 9.81 * 0.0009975 * 35.9875(7715 - 965)}{9 * 0.55})^2 * 0.0000016073^2 + \\ (\frac{2 * 9.81 * 0.0009975^2 * 35.9875}{9 * 0.55})^2 * 37.9363^2 + (\frac{2 * 9.81 * 0.0009975^2 * 35.9875}{9 * 0.55})^2 * 1.443376^2 \\ + (\frac{2 * 9.81 * 0.0009975^2 (7715 - 965)}{9 * 0.55})^2 * 0.169918^2 + \\ (\frac{-2 * 9.81 * 0.0009975^2 * 35.9875(7715 - 965)}{9 * 0.55^2})^2 * 0.000289^2\end{array}}$$
$$u(\eta) = \sqrt{\begin{array}{l}0.000009531871942515 + 0.0000289902875797454 + 0.00000004196638309775 + \\ 0.000020460938337593788 + 0.00000025340806554346\end{array}}$$
$$u(\eta) = 0.0077 N * s * m^{-2}$$
$$----$$
$$\eta = \mathbf{(0.9580 \pm 0.0077) N * s * m^{-2}}$$